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Instantaneous hydrolysis of PET bottles: an efficient pathway for the chemical recycling of condensation polymers†

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Finding ways to restore the environmental balance that has been disturbed by the rise in the use of plastic commodities is urgent for researchers and the wider community. It is imperative that the increasing amount of plastic waste and the high amount of petrochemical resources consumed during the constant replacement of single-use plastics be reduced. Poly(ethylene terephthalate) (PET) is one of the most commonly produced single-use polymers in the world, and its mechanical recycling is challenging due to the loss of properties during reprocessing. Chemical recycling is a feasible alternative to reclaim the monomers, however, its viability relies on establishing a straightforward, fast, and inexpensive procedure to turn the end-of-use polymer into new pure monomers. This work reports on the fastest known procedure for PET chemical recycling to produce terephthalic acid and ethylene glycol in an efficient and straightforward manner, thanks to microwave-assisted heating that permitted 100% PET conversion into TPA in just 1 minute at 120 °C. The depolymerization kinetics of this new procedure were studied and its improved efficiency over other reported hydrolysis procedures was attributed to a thicker shrinking layer. This new procedure may form a major breakthrough in chemical depolymerization. Evidence pointing to the higher reactivity of free OH species enabled us also to obtain dimethyl terephthalate (DMT) through the use of an anhydrous depolymerization system that is able to convert PET into DMT in 4 min at 80 °C, but requiring careful humidity control. The activation energy for the proposed depolymerization system was estimated as 120 kJ mol⁻¹. An additional advantage of the proposed process is the production of potassium sulfate as a side-product, a valuable fertilizer with high market value. This product compensates for the lack of recoverability of the catalyst. Our KOH-in-methanol hydrolysis (KMH) process has the potential to become a widely used depolymerization solution for a wide range of (heterogeneous) condensation polymers, which is currently under study by our research group.

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1. Introduction

Poly(ethylene terephthalate) is one of the most consumed polymers in the world. While leading to a large reduction in CO₂ emissions from manufacturing and weight reduction during transport over glass, its use has also led to significant, and growing, environmental issues coming to the fore. Contamination of marine and land ecosystems by micro and macro plastics is an urgent threat, as it has been reported that microplastics are present in at least 12 species of fish in the North Sea, of which at least 3% is PET.¹ Not only animals are affected by micro-plastics in the ecosystem, drinking water can

also be affected as demonstrated recently.² These problems arise from inadequate disposal of plastic waste, which to a significant extent is caused by poorly managed recycling. Recently, a magnifying effect has been the accelerated increase in the consumption of single-use plastics for personal protection, due to the COVID-19 pandemic.³ Despite increased awareness about selective collection of plastics and the availability of a growing number of biodegradable plastics,⁴ the reduction of the environmental impact of plastic usage in society remains an important challenge. Different ways in which plastics can be recycled include re-use, mechanical recycling (re-processing), energy recovery (burning the plastics to use the energy), or chemical recycling (reclaiming the original components of the polymer). Mechanical recycling is the traditional pathway for plastic recycling, however, it tends to involve significant loss of properties, triggered by chain scission or degradation during re-melting, and removal of additives may be complex or not cost-effective. Reduction in the

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mechanical performance makes it necessary to incorporate fresh feedstock in order to achieve the desired properties for the new products prepared from the recycled polymer.⁵ Chemical recycling of polymers breaks the polymer back into its constituents (or other precursors), making it possible to separate the monomers and remove additives by traditional separation techniques, after which a new polymer can be produced, indistinguishable from the non-recycled polymer.

Chemical recycling of PET can be performed *via* nucleophilic or electrophilic attack on the carbonyl by several different agents. This kind of process can render monomers for PET production or terephthalamides (for the synthesis of other high value-added materials). The advantages of chemical recycling include the reduction in oil feedstock consumption, along with its versatility to produce high value-added materials, the possibility of producing new polymers with the same properties of the ones produced using petrochemical feedstock, and relative ease of removal of cumbersome additives and colorants. Life cycle assessments have previously demonstrated that chemical recycling of plastics offers high potential for reducing CO₂ emissions compared to mechanical recycling, based on the quality of the obtained products that can directly be used to produce virgin-like materials, while mechanical recycling offers, most of the time, downgraded pellets that underperform the original materials, making a constant input of new petroleum-based monomers necessary. Furthermore, it is expected that a chemical recycling process involving low or zero organic solvents, will increase the energy economy of the process and consequently diminish the environmental impact of a recycling plant.⁶

Hydrolysis is the chemical reaction that produces chain scission in the PET chain, producing terephthalic acid (TPA) and

ethylene glycol (EG) as final products (Fig. 1). This reaction can be achieved either by acidic or basic aqueous processes. Many reports are available in the literature, reporting on a wide variety of different depolymerization conditions, covering inorganic acids to alkaline hydroxides, phase transfer catalysts, as well as microwave heating and ultrasound. Table 1 summarizes a selection of important works and conditions found in the literature for the hydrolysis of PET,⁷ including the values of environmental factors calculated following the description in Section 2.2.1. One can see that on average, using aqueous solutions of alkaline hydroxides or concentrated acids, 100% PET conversion is only possible at temperatures above 100 °C and for several hours (which can be brought down to 1–2 h by incorporating a phase transfer catalyst).

The aqueous hydrolysis of PET is a heterogeneous reaction between two incompatible phases: a solid (PET) suspended in an aqueous phase. One of the biggest issues for hydrolysis in aqueous media is the incompatibility between the liquid phase and the polymer. Consequently, the incorporation of a phase transfer catalyst has proved to be an important strategy to improve the reaction efficiency, as observed by several research groups.^{10,14} An alternative way to improve reaction efficiency includes the use of microwaves, which reduces the environmental energy impact by increasing heating efficiency, thereby diminishing the reaction time.¹² The present work explores the combined use of microwave irradiation and a solution of potassium hydroxide in anhydrous alcohol to develop an energy efficient non-aqueous hydrolysis process. We believe that the *in situ* formed potassium methoxide will have improved diffusivity into the polymer matrix further speeding up the depolymerization reaction. The viability of the proposed method is completed by the formation of potassium sulfate, the byproduct generated by neutralization of the reaction mixture. Potassium sulfate is an important fertilizer used to provide sulfur and potassium to the soil.¹⁵ Hence, the proposed process has two valuable outputs that have the potential to be a game changer in polymer recycling.

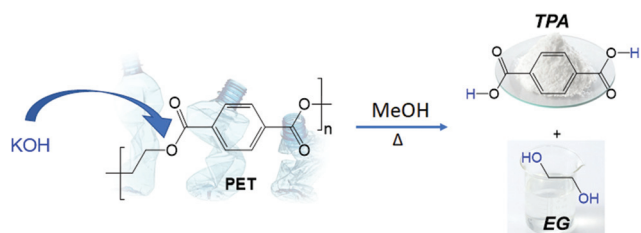


Fig. 1 Illustration of chemical depolymerization of PET *via* hydrolysis of ester bonds by acid or basic catalysts.

2 Materials and methods

2.1 Materials

Methanol (>99.9%) was provided by Fischer Scientific. Transparent PET bottles were cut in rectangular flakes of

Table 1 Summary of the most outstanding PET hydrolysis processes found in the literature for complete depolymerization of PET

Ref.	Catalyst	Co-catalyst/conditions	<i>T</i> (°C)	Reaction time (min)	<i>E</i> factor (a.u)	ϵ (°C ⁻¹ min ⁻¹)	ξ (°C min)
8	H ₂ SO ₄ (aq)	N.R.	135	300	887	222×10^{-5}	399×10^5
9	HNO ₃	DMSO Na ₂ SO ₄	98	140	586	507×10^{-5}	116×10^5
9	HNO ₃	DMSO Na ₂ SO ₄	110	140	533	497×10^{-5}	107×10^5
9	HNO ₃	DMSO Na ₂ SO ₄	120	140	468	519×10^{-5}	901×10^4
10	NaOH (aq)	TBAI	90	45	39	244×10^{-4}	160×10^4
11	NaOH (aq)	EtOH : water 80 : 20	110	30	1228	270×10^{-4}	455×10^4
12	NaOH (aq)	Microwave	220	2	172	220×10^{-3}	780×10^2
13	HNO ₃	DMSO Na ₂ SO ₄ H ₃ PO ₄	140	160	56	437×10^{-5}	128×10^5

around 0.5 cm², washed with soap and distilled water, and dried under vacuum before the reaction. Potassium and sodium hydroxide were provided by VWR Chemicals (89.3%).

2.2 Methods

2.2.1 Synthesis of terephthalic acid from PET waste. Approximately 0.5 g of the cut PET flakes was charged into a high-pressure vial along with 10 ml of the KOH-in-methanol (KMH) solution. The vial was closed with a hermetically sealed metallic lid and placed into the microwave reactor (Initiator+ Microwave system, Biotage, Sweden). Power was supplied in pulses until a stable temperature response was achieved. The reaction time was taken from the moment the system reached 5 °C below the programmed temperature. The system was stirred magnetically at 600 rpm. After the reaction time, 10 ml of distilled water was added to dilute the “salted out” potassium methoxide that precipitated along with terephthalate salts. The insoluble unreacted PET was filtered off, washed with distilled water (200 ml), dried under vacuum at 80 °C, and weighed. Acid addition was stopped around pH 4. The white solid was filtered off, washed with distilled water (200 ml) and methanol, dried under vacuum, and weighed.

PET conversion was calculated following eqn (1), where $m_{\text{un.PET}}$ is the mass of the unreacted polymer and m_{PET} is the mass of the initial polymer. TPA yield was calculated using eqn (2), where m_{TPA} is the mass of TPA obtained after the reaction. The 0.874 coefficient corresponds to the ratio of molar mass of TPA and the molar mass of PET-mere. The balance for mass determination of PET before and after the reaction had a resolution of ± 0.0001 g.

$$\text{PET conversion (\%)} = \frac{m_{\text{un.PET}}}{m_{\text{PET}}} \times 100 \quad (1)$$

$$\text{TPA yield (\%)} = \frac{m_{\text{TPA}}}{m_{\text{PET}} \times 0.874} \times 100. \quad (2)$$

2.2.2 Production of dimethyl terephthalate. In an oven-dried high-pressure vial, approximately 2.48 g of metallic sodium was left to react with methanol under an argon flux. Once the sodium was consumed, approximately 0.5 g of PET was added and the system was sealed and further taken to the microwave reactor. After the reaction time, 10 ml of distilled water was added. The remaining white solid was filtered off, dried and weighed, while the remaining solution was neutralized to pH 4 and processed as described in Section 2.2.1.

2.2.3 Green chemistry calculations. The energy efficiency coefficient ϵ is a way to directly compare the proposed method with others to verify its efficiency, and it is defined by eqn (3) where t is the reaction time (in minutes), T the reaction temperature in degrees Celsius, and Y the yield of the main monomer as a mass fraction.⁷ The lower the temperature and the reaction time, the higher the relative value of ϵ . The E factor is based on the environmental factor (E), proposed by Prof. Sheldon in 1992,¹⁶ which has been used in green chemistry metrics with success¹⁷ and corresponds to a simple mathematical relation defining the amount of waste generated per

unit of the product. In this work, we propose a modification of the E factor, considering that it has been established that 90% of solvent can be recycled in industrial processes,¹⁸ leading to eqn (4), used to compare the E factor of different PET depolymerization works in the literature and the one presented here. The environmental energy impact corresponds to the combined effect of the mass input and the energy economy as described in eqn (5).

$$\epsilon = \frac{Y}{T \times t} \quad (3)$$

$$E_{\text{factor}} = \frac{\left[0.1 \times \left(\frac{\text{solvent}}{\text{PET}} \text{ ratio} \right) + \left(\frac{\text{cat}}{\text{PET}} \text{ ratio} \right) + \left(\frac{\text{other subst}}{\text{PET}} \text{ ratio} \right) \right] \times m_{\text{PET}}}{\text{Yield}_{\text{Product}} \times \frac{\text{MM}_{\text{Product}}}{\text{MM}_{\text{PET MERE}}} \times m_{\text{PET}}} \quad (4)$$

$$\xi = \frac{E_{\text{factor}}}{\epsilon}. \quad (5)$$

2.2.3 Characterization

2.2.3.1 Fourier transform infrared (FTIR) spectroscopy. FTIR analyses were performed on an Alpha 1 spectrophotometer (Bruker) operated in attenuated total reflection mode with single reflection on unreacted polymers and depolymerization products combining 24 scans between wavenumbers 450 and 4000 cm⁻¹.

2.2.3.2 Thermogravimetric analysis (TGA). TGA was performed on a Netzsch Tarsus TG209F3 using platinum pans for the polymers and alumina pans for the depolymerization product. The apparatus was equipped with a differential thermal accessory for the determination of thermal transitions. Amounts ranging from 5–10 mg were loaded in the pans and the analyses were carried out using air as the carrier gas and nitrogen as the protective flow gas for the microbalance. A heating rate of 10 °C min⁻¹ was used from 30 to 900 °C.

2.2.3.3 Differential scanning calorimetry (DSC). DSC analyses were carried out on a DSCQ2000 (TA instruments) using aluminum Tzero pans. The analysis consisted of heating the polymer from room temperature to 280 °C, keeping the sample isothermal for 5 min followed by rapid quenching to -60 °C, aiming to amplify the signal for the amorphous region and to promote cold crystallization. After stabilization at -60 °C, the sample was heated at 5 °C min⁻¹ to 280 °C to determine the glass transition temperature (T_g), cold crystallization temperature (T_{cc}), melting temperature (T_m) and enthalpies of cold crystallization and melting (ΔH_{cc} and ΔH_m). For the analysis of the depolymerization product, a single heating cycle from room temperature to 280 °C at 5 °C min⁻¹ was applied.

2.2.3.4 Nuclear magnetic resonance (NMR) spectroscopy. The ¹H-NMR spectroscopy analyses were carried out on a Spinsolve 60 Ultra (Magritek). The analyses were carried out on products at a concentration of 20 mg ml⁻¹ in deuterated DMSO as the solvent.

3 Results and discussion

3.1 Depolymerization using conventional heating

Following the work by other authors in previous studies of adding alcohol in basic hydrolysis media,^{11,19,20} we performed a comparison of the depolymerization of PET by basic solutions of potassium hydroxide in methanol, ethanol, iso-propanol, and *tert*-butanol in the absence of water. The present method differs greatly from the approach of pre-soaking PET in organic solvents,²¹ making the process more sustainable and straightforward as only alcohol and alkali are used. The product for all the reactions is TPA as demonstrated by FTIR spectroscopy (Fig. 2, the details of the absorption band attribution can be found in Table S1†). No differences were found either among the depolymerization products or the PET conversion or TPA yield for the alcohols studied. This indicates that the alcohol itself is not the reactive species but rather acts as a medium for the alkaline catalyst, which is reinforced by the fact that the amount of acid needed for neutralization of the reaction medium does not depend on the PET conversion. The efficiency of the alcoholic medium for the hydrolysis of PET under basic conditions would be a function of its capacity to dissolve the highest amount of base, since it is a known fact exposed in multiple studies that the reaction speed increased and highest conversions are reached when the base concentration increases.^{10–12,14} Taking this into consideration, further reactions were carried out in methanol, since it could dissolve the highest amount of base.

The effect of the base cation was studied by comparing the performance of KOH and NaOH in methanol. Fig. 3a shows PET conversion as a function of reaction time for analogous reactions using NaOH and KOH keeping all the other variables constant. The reaction was significantly faster when KOH was employed, which could be attributed to the stability of the KMH solution compared to the low stability of NaOH in solution, a fact that is evidenced by the precipitation of NaOH at high concentrations in methanol, which is not observed in methanolic solutions of KOH.

We can compare the efficiency of the KMH solution process used here with the aqueous hydrolysis processes in published work:^{10,11} The traditional aqueous hydrolysis method took 50 h at 110 °C to be completed without the incorporation of a phase transfer catalyst, while with use of 2 wt% of a phase transfer catalyst 100% conversion of PET was achieved in 5 h. This comparison demonstrates the efficiency of the KMH system, which is faster and requires less energy than the aqueous approaches. The efficiency of the KMH system was also proved in a room temperature reaction (Fig. 3c), as PET was fully depolymerized under stirring in 3 days. This performance of the KMH system performs beyond that of other depolymerization studies at room temperature irrespective of the solvent and catalyst system used.²²

3.2 Depolymerization using a microwave reactor

PET conversion as a function of the base concentration was monitored in order to verify the optimal concentration at which microwave depolymerization could be performed with maximum energy efficiency. Fig. 4 shows PET conversion as a function of the molarity of the depolymerization solution for a reaction period of 3 min at 80 °C. PET conversion increased with increasing KOH in MeOH concentration up to 1.25 mol L⁻¹, where it reached a plateau. Increasing the KOH concentration further did not lead to a further increase in PET conversion; it even showed a tendency to decrease. This behavior suggests an active participation of KOH in the reaction mechanism, yet, it was verified that KOH is not consumed during the reaction, as the amount of acid required for neutralizing the reaction mixture remains unaltered regardless of the amount of PET converted. The behavior of PET conversion as a function of the molarity of KOH in the depolymerization solution can be explained by the saturation of the solution with the formed potassium terephthalate. The produced salt requires more methanol molecules to be solvated, as a result, the remaining methanol is unable to solvate all the solute, which causes the base to salt out of the solution along with terephthalate, leading to an actual decrease of base concentration

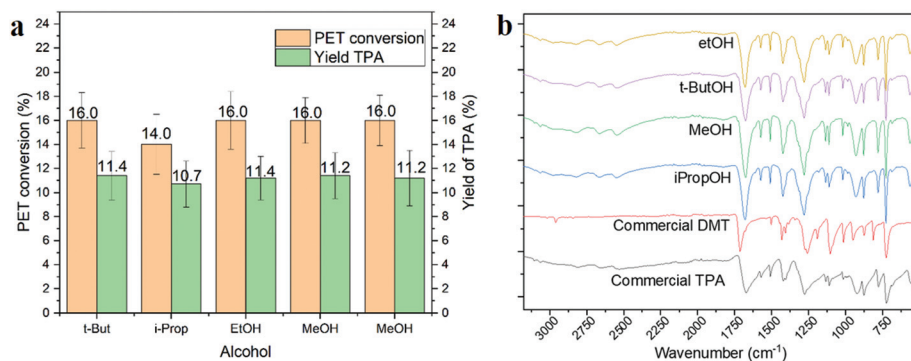


Fig. 2 (a) PET conversion and TPA yield as a function of alcohol for a series of analogous reactions where the KOH solution was prepared in *t*-butanol, isopropanol, ethanol, and methanol. The reactions were performed under an air atmosphere using a 0.25 M concentration of base (the experiment marked with * refers to a reaction under an argon atmosphere). (b) FTIR spectra of the depolymerization products compared to those of commercial TPA and DMT.

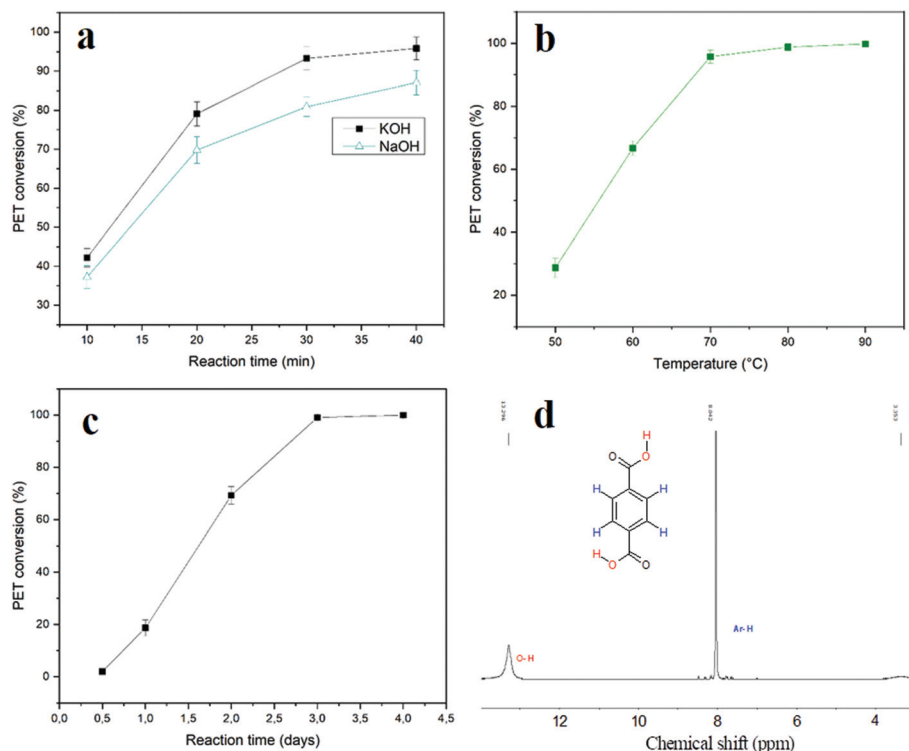


Fig. 3 (a) PET conversion as a function of reaction time in 1.25 M solutions of KOH and NaOH in methanol (PET : solution 1 : 20 g : ml) at 80 °C. (b) PET conversion as a function of temperature for 1.25 M KOH/MeOH solutions (PET : solution 1 : 20 g : ml). (c) PET conversion (%) as a function of reaction time at room temperature (PET : solution 1 : 20 g : ml). Representative NMR spectrum of the obtained TPA solid.

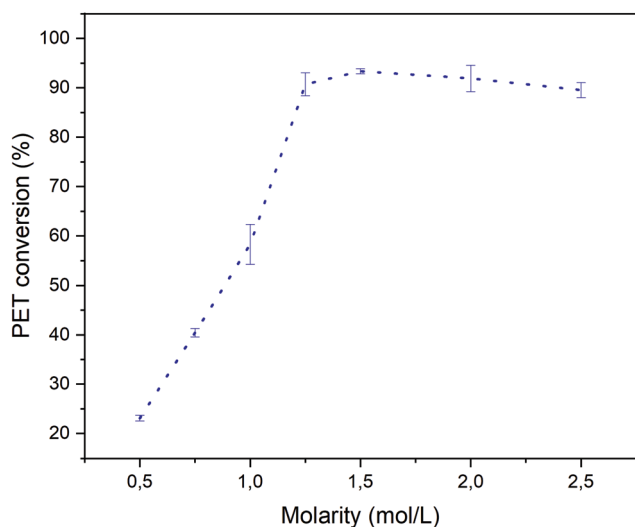


Fig. 4 PET conversion as a function of molarity of KMH solution. Experiments were carried out at 80 °C for 3 min and PET : solution ratio of 0.5 g : 10 ml.

in the liquid phase with PET conversion, decelerating the reaction and thus, causing the plateau observed in Fig. 4. We confirmed this by modifying the procedure described in Section 2.2.1, adding a filtration step of the reaction mixture after the reaction period, which permitted to verify that the basicity of

the filtered solution diminished along with PET conversion. This finding does not mean that KOH is actually consumed, it means that it leaves the liquid phase, causing the observed deceleration of the reaction. Given the apparent maximum for PET conversion around 1.25 mol L^{-1} KOH in MeOH, further experiments were carried out at this concentration.

Optimization of the reaction time for different temperatures is presented in Fig. 5, where it is observed that PET conversion increased with the reaction time for all investigated temperatures. Under microwave reactor conditions, all temperatures studied corresponded to an exponential increase of pressure, which is likely one of the factors responsible for the significant reduction in the required reaction time to attain 100% PET conversion with increasing temperature. When the reaction temperature was set to 120 °C, only 1 min was required for complete PET conversion into soluble potassium terephthalate. Upon addition of water after 1 minute of reaction time, a clear homogeneous solution was observed as the precipitated base was dissolved. On the other hand, Fig. S9† depicts the evolution of PET conversion as a function of temperature for different reaction times under the same reaction conditions as before. It was observed that the minimum time and temperature necessary for 100% conversion of PET flakes are 4 min at 100 °C.

3.2.1 Reaction kinetics. The activation energy for the depolymerization reaction of PET flakes was calculated by plotting the inverse of the concentration of poly(ethylene terephthalate)

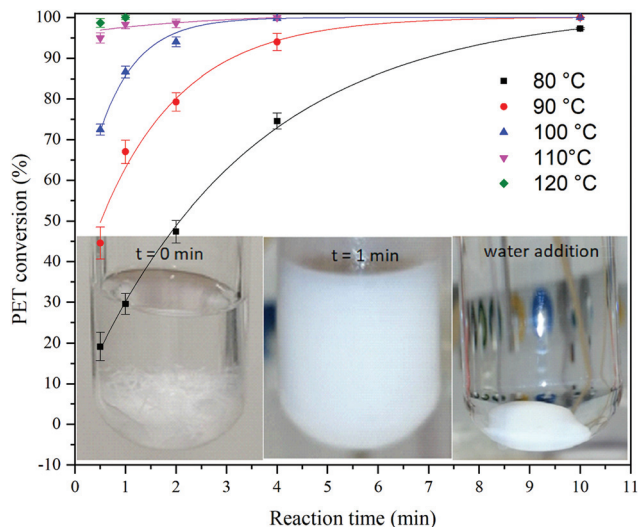


Fig. 5 PET conversion as a function of reaction time for a microwave reactor with PET:solution ratio of 0.5 g:10 ml and 1.25 M KOH in methanol at different temperatures. The inset image shows the reaction progress for 1 min at 120 °C: before reaction, showing the PET flakes; and after 1 min of reaction showing the “salted out” potassium methoxide and after water addition, where a transparent homogeneous mixture is observed.

in the reaction system in (mol L^{-1}) as a function of the reaction time. Working at four different temperatures (65, 80, 90 and 100 °C), four different slopes were obtained, which correspond to the reaction rates. Applying the Arrhenius eqn (6), the activation energy for the depolymerization reaction of polyethylene terephthalate pellets using the developed KMH system was estimated to be $120.7 \text{ kJ mol}^{-1}$ (Fig. 6B). This activation energy is in agreement with values reported for conventional hydrolysis systems, which indicates that the combination of microwaves and the developed high pressure inside closed reaction vessels can be the cause of the outstanding performance of the proposed system.²³

$$k = Ae^{-E_a/RT} \quad (6)$$

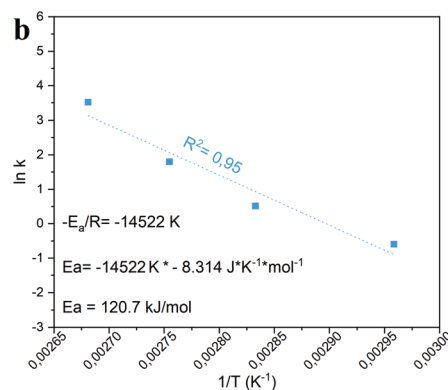
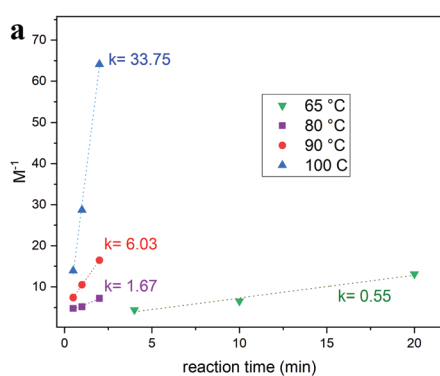


Fig. 6 (A) Reciprocal of the concentration of polyethylene terephthalate as a function of reaction time at 65, 80, 90, and 100 °C. (B) The Arrhenius plot for the four temperatures studied in the depolymerization reaction of polyethylene terephthalate pellets in the KMH system.

$$\ln(k) = -\frac{E_a}{R} \left(\frac{1}{T} \right) + \ln(A). \quad (7)$$

3.2.2 Reaction mechanism. The outstanding results reported in this work require a deeper explanation of the chemistry involved in the reaction. Since the KOH-in-methanol solution produces potassium methoxide in an equilibrium reaction, it would be expected that dimethyl terephthalate would have been produced along with terephthalic acid. However, the evidence showed that for all the temperatures studied, the depolymerization using KMH solution, produced exclusively TPA as the depolymerization product, which is a hydrolysis product, instead of DMT, which is the methanolysis product. DSC, FTIR and NMR, demonstrated that TPA is indeed the sole product of this depolymerization process without any DMT being formed. Fig. 7 shows a comparison between the proposed mechanism for the depolymerization of PET by the KMH solution and the model study in which metallic sodium is dissolved in methanol, generating anhydrous sodium methoxide *in situ*. The depolymerization by Na^0/MeOH should lead exclusively to the methanolysis product: dimethyl terephthalate. We propose that the difference in the reactivity for methanolysis and hydrolysis is explained by the higher mobility of KOH compared to potassium methoxide, which makes this reaction temperature-driven (while methanolysis would be diffusion-driven). In order to prove this hypothesis, reactions of the Na^0/MeOH system were performed on PET pellets using 99.9% pure methanol for 5 min, keeping the concentration and PET:solution proportion approximately constant (Fig. 8). The experiments showed a clear competition between methanolysis and hydrolysis, where high temperatures clearly favor hydrolysis over methanolysis, while there is a tendency for methanolysis prevalence at lower temperatures. This trend corroborates that free OH, coming from the interaction of water traces in methanol and the atmosphere, has a high mobility and reacts preferentially at higher temperatures over methoxide. Thus, the nucleophilic attack of methoxide appears to be diffusion driven, while the hydroxide attack is temperature driven. In order to support this hypothesis

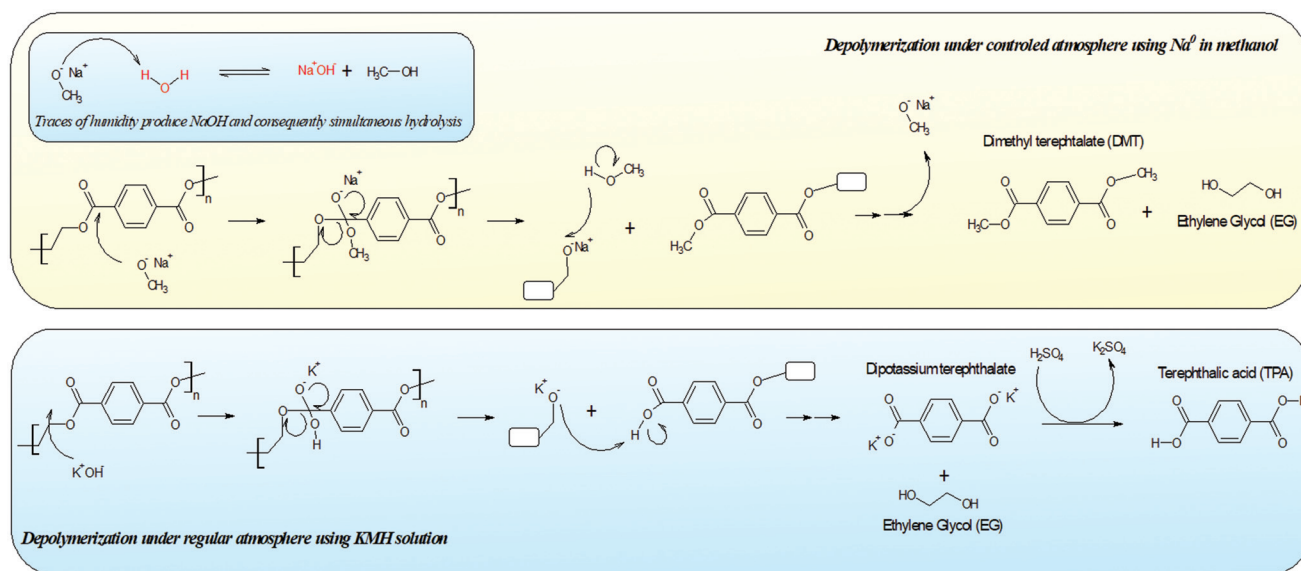


Fig. 7 Proposed chemical reaction mechanism for the depolymerization of PET using the KMH solution and an analogous system using metallic sodium in methanol. The inset scheme shows the reaction of water traces present in the solvent and the *in situ* formed sodium hydroxide.

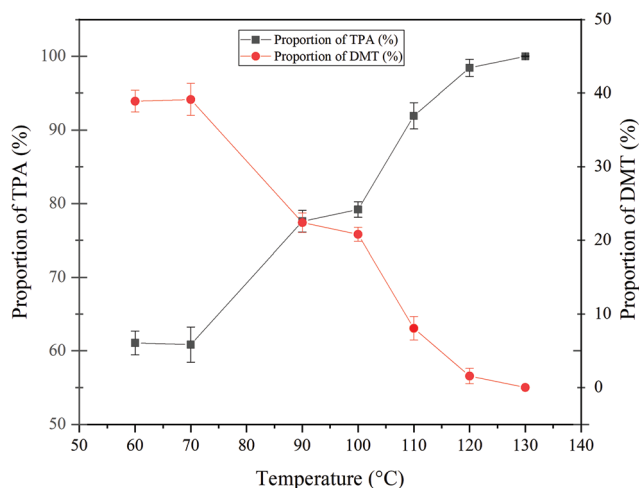


Fig. 8 Proportion of the produced TPA and DMT as a function of temperature for a competitive reaction of methanolysis-hydrolysis using the sodium/methanol system.

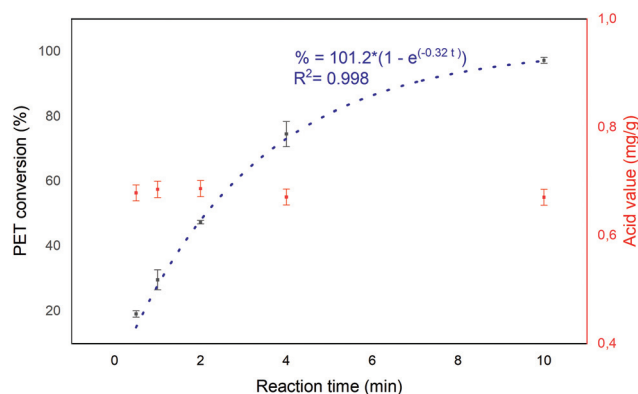


Fig. 9 PET conversion and acid value of the obtained TPA as a function of the reaction time for KMH depolymerization at 120 °C, 1.25 M and 0.5 g : 10 ml PET : KMH ratio. The exponential equation that fits the experimental points of the curve is shown ($R^2 = 0.998$).

further, a reaction was carried out overnight at room temperature and it produced nearly 100% DMT. The same system was tested at 80 °C using methanol dried over molecular sieves, which produced DMT in more than 80% yield. We believe that further control over the reaction conditions can lead to a near 100% DMT yield.

3.3 Interaction of phases and reaction velocity

The PET conversion and acid values of the obtained product as a function of the reaction time for the chosen KMH system (80 °C, 1.25 M, 0.5 g : 10 ml) is shown in Fig. 9. The virtually constant acid value of the TPA produced throughout the duration of the reaction, combined with the FTIR spectra (see the

ESI⁺), showed that the only soluble product derived from the process is the TPAK₂ salt. After chain scission in the solid matrix, the shortened chains seem to remain in the solid state so that no other insoluble product is observed. This implies that there is a chain shortening and oligomers can be found at some point during the depolymerization process, however, the speed of the depolymerization is such that the reaction proceeds almost instantaneously to the TPAK₂ salt. Consequently, what limits the reaction speed is the size and shape of the polymer fed to the reactor (additional experiments showed that big PET pellets take a longer time to fully depolymerize). The unreacted polymer, by the end of the reaction period, is affected only to a limited extent (*vide infra*), which results in a slight molar mass loss and increase in crystallinity, a consequence of the preferential attack of the depolymerization reac-

tion in the amorphous regions. The only products are TPAK₂ (soluble) and unreacted PET (insoluble).

The core-shrinking model²⁴ was used as a basis to explain the high velocity observed in the depolymerization reactions studied here (Fig. 10). The comparative description of the aqueous and non-aqueous media points to the possible effect of the compatibility between the reactants and the reaction medium on the velocity of the reaction. In the case of the non-aqueous system, the higher compatibility (translated as the higher solubility parameters encountered for PET/MeOH than for PET/water) promotes the formation of a wider shrinking layer which, as a result, enables a higher reaction rate in the reaction system, responsible for the outstanding results observed for the KMH system under its ideal conditions. The conversion of PET into TPAK₂ salt is an irreversible process since the dipotassium salt cannot undergo an esterification reaction with EG, methanol or even potassium methoxide as the carbonyl carbon is insufficiently electrophilic.²⁵ It is

important to highlight that the difference in the solubility parameter is not the only factor that explains the outstanding results but they are due to a combination of compatibility (augmented by the presence of a base, which can interact with the carbonyls along the polymer chain) and basicity.

As observed by multiple researchers in previous work on the solvolytic depolymerization of PET,^{26–29} chain scission occurs preferentially in the amorphous regions of PET, as confirmed by the increase in the melting enthalpies of the unreacted PET with increasing reaction time (Fig. 11A). This indicates that the crystalline regions are retained longer than the amorphous ones, as the amorphous regions are more rapidly attacked and converted into soluble products. The DSC plots also show another particular fact: the marked inhibition of cold crystallization at around 30 s reaction time, attributed to chain shortening that occurs at low reaction times. This is corroborated by differential thermogravimetric analysis (Fig. 11B) showing a constant shift of the $T_{\max}$ of the main

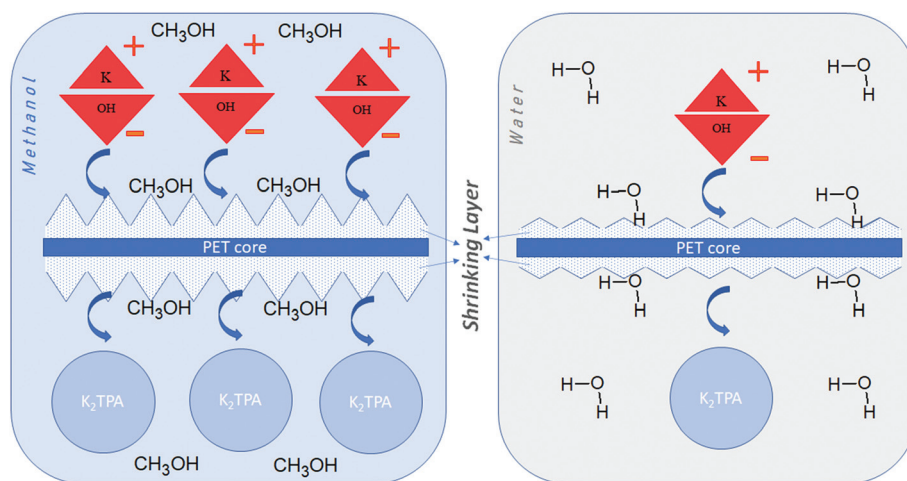


Fig. 10 Interpretation of the differences observed between the reaction in aqueous and methanolic media based on the core-shrinking model of the KMH depolymerization system over PET flakes.

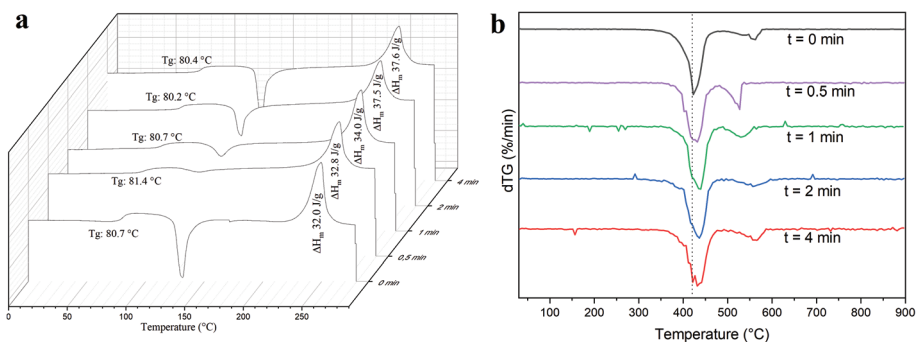


Fig. 11 (A) DSC plots of unreacted PET submitted to KMH depolymerization at 80 °C for different reaction times ranging from 0 to 4 min in the microwave reactor, keeping the PET : KMH solution ratio at 0.5 g : 10 ml, 1.25 M. DSC analysis was performed at a heating rate of 5 °C min⁻¹. Only the second heating cycle after quenching at –80 °C is shown. (B) Differential thermogravimetric analysis for the unreacted PET subjected to KMH depolymerization at 80 °C for different reaction times ranging from 0 to 4 min in the microwave reactor, keeping the PET : KMH solution ratio at 0.5 g : 10 ml, 1.25 M.

degradation step to higher temperatures. Hence, the more crystalline regions (with higher degradation temperature) become more prevalent inside the matrix as the reaction time increases.

3.3.1 Effect of water on reaction efficiency. The presence of additional water in the reaction system has a negative impact on the performance of the KMH solution. When increasing the water content up to 2.5% in the KMH solution under ideal conditions to achieve 100% PET conversion (1 min, 120 °C), it was seen that PET conversion decreased when the water content surpassed 1% (Fig. 12). Therefore, it is important to use pure methanol and limit the water content to below 1% in the KMH solution. Similar results were found by Pham and Cho (2021),³⁰ investigating methanolysis of PET catalyzed by potassium carbonate. They determined that excess water (molar ratio H₂O:PET around 0.8) was able to noticeably decrease PET conversion and even promote the competition between methanolysis and hydrolysis. In the present work, as

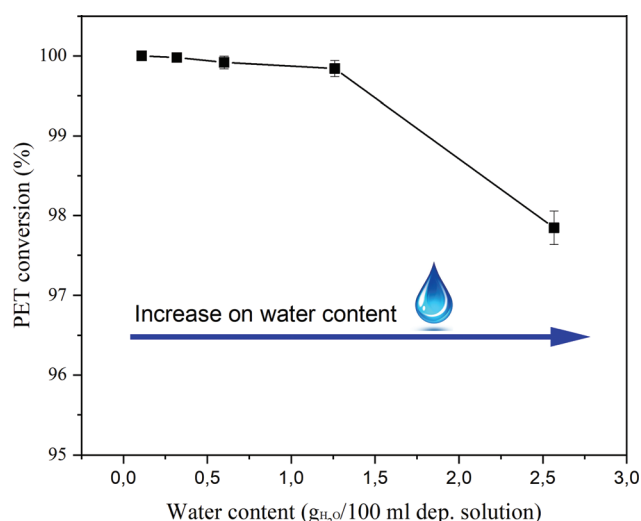


Fig. 12 PET conversion as a function of the water content of the KMH solution in a reaction mixture containing 0.5 g:10 ml PET:KMH (molarity of KMH solution: 1.25 mol L⁻¹ at 120 °C for 2 min).

previously explained, the competition is favorable to the hydrolysis product, obtaining exclusively TPA after work up.

The effect of water can be thus be explained using the aforementioned core shrinking model: increasing water content affects the solubilization of the outer layers and causes a reduction of the reactive shell volume thereby diminishing the reaction velocity and thus the PET conversion for water concentrations above 1%.

3.4 Industrial feasibility and green chemistry metrics

Five g of PET flakes were then reacted to verify the inputs and outputs for a hypothetical industrial process of PET depolymerization with the KMH system as illustrated in Fig. 13, where all the inputs and outputs are displayed as a projection based on the results obtained in this work (the yields are expected to be significantly higher on an industrial scale due to reduced losses during purification). Potassium sulfate is a commonly used salt in fertilizers, providing sulfur and potassium to the soil. Therefore, the present process is able to produce two useful products out of PET waste. The yield of terephthalic acid was 93.8% and its purity was confirmed by DSC, FTIR, TGA, and NMR spectroscopy (see the ESI†). The TPA yield obtained in this work was used for the calculation of the green chemistry metrics as defined in Section 2.2.2. Fig. 14 presents a comparison among the three green chemistry parameters, namely *E*, *ε* and *ξ* for different literature reports dealing with PET depolymerization using microwaves. A great difference in *ε* factor is observed for the present work, which exceeds by at least 6 times the best work found in the literature. Paliwal and Mungray¹⁰ used significantly higher temperatures and employed a phase transfer catalyst, a drawback in terms of green chemistry metrics. On the other hand, Căta and co-workers¹² employed significantly higher temperatures than the ones employed in the present study, which is also reflected in the green chemistry metrics. The positive chemistry metrics for the KMH systems is a decisive step forward towards the application of a feasible industrial process for chemical recycling of PET, since it is fast, straightforward and energetically favorable.

Other works in the literature claim that the use of heterogeneous catalysts is a decisive factor for the implementation of

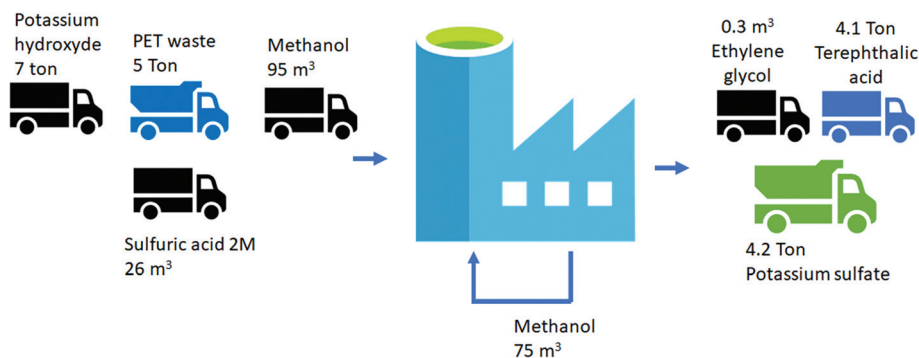


Fig. 13 Scaling up of PET depolymerization by the KMH system and the inputs of a hypothetical industrial process.

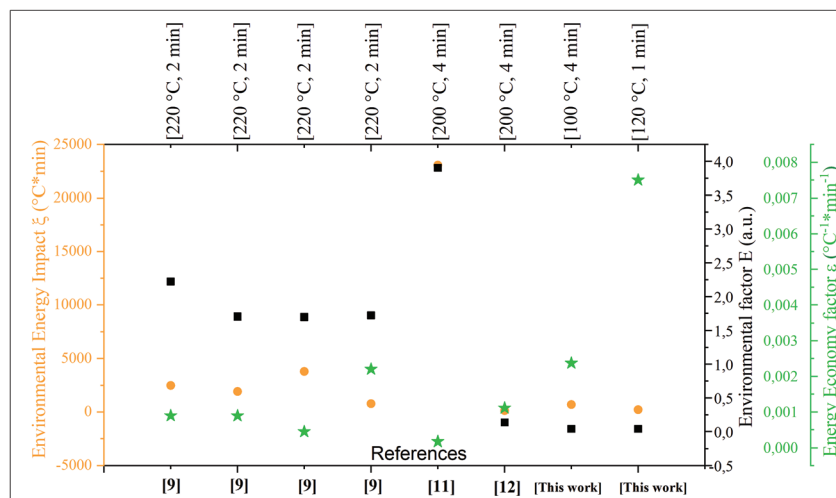


Fig. 14 Green chemistry metrics for PET depolymerization processes found in the literature using microwave. In black, the environmental factor, in green the energy economy factor, and in orange the environmental energy impact.

chemical recycling, given the possibility of easy catalyst recovery.^{31,32} However, in our work, the catalyst is converted into potassium sulfate, a valuable byproduct in its own right that could cover the lack of catalyst recovery. The ease of the current process is such that it has the potential to be applied directly in small factories close to waste generating communities. One evident environmental advantage that should make a decisive change in current chemical recycling processes is the avoidance of organic solvents such as chloroform, dichloromethane or tetrahydrofuran, making the KMH solution not only less energetically demanding, but also less harmful to the environment. This could be especially beneficial in places with environmental plastic waste problems, such as Karachi, Pakistan, a city with a serious environmental issue related to the high amount of plastics in rivers and beaches, directly affecting the quality of drinking water and the soil.^{33,34} The development of recycling opportunities that can be implemented within local communities and the use local plastic waste as the resource could lead to successful management of the local environmental issues while also providing economic benefits.

Since plastic waste is normally heterogeneous, the present process presents important advantages with respect to mechanical recycling, since the solubility of most plasticizers will permit a straightforward separation of contaminants from the desired monomers, while in a mechanical recycling process, the issue of molar mass loss will be accompanied by the presence of residual contaminants. Currently, our group is working on a straightforward process that uses KMH for mixed streams of polycondensation plastics.

4. Outlook and conclusions

The present work gives an insight into the properties of the KMH system for almost instantaneous depolymerization of

PET. The lowest time required for complete depolymerization was 1 min at 120 °C, while the minimum temperature at which complete depolymerization was achieved was 100 °C for 4 min of reaction time. Both results are breakthroughs compared to the state of the art and represent a feasible option for implementation in recycling plants around the world, due to the simplicity and energy efficiency of the process, as well as the production of a fertilizer as a by-product. A swollen shrinking layer in a core shrinking model was proposed to explain the high rate of the reactions carried out with the KMH system. The higher compatibility between the PET matrix and the depolymerization solution is believed to create a wider reactive layer, which favors a rapid interaction between the base and the polymer. It was verified that amorphous regions are more susceptible to reaction than the crystalline ones as verified by DSC and DTGA. The activation energy for the depolymerization reaction was estimated as 120.7 kJ mol⁻¹. The effect of water on the depolymerization reaction was evaluated and revealed that the system performs well up to 1% water content, after which PET conversion suffers a significant decrease, believed to be due to a reduced surface of the reactive outer layer and consequently the rate of depolymerization.

Conflicts of interest

There are no conflicts to declare.

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